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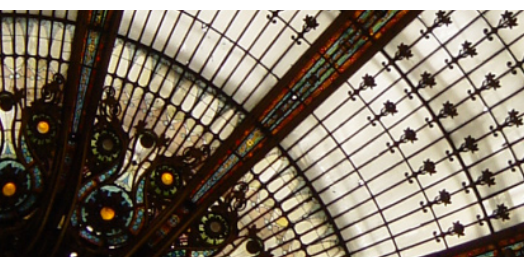
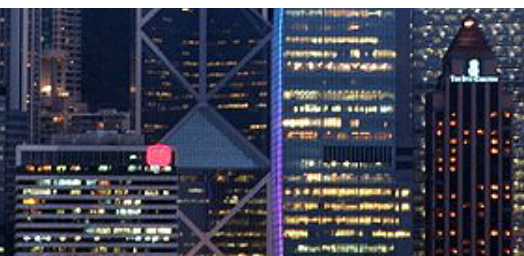
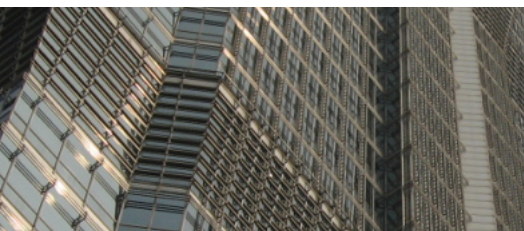
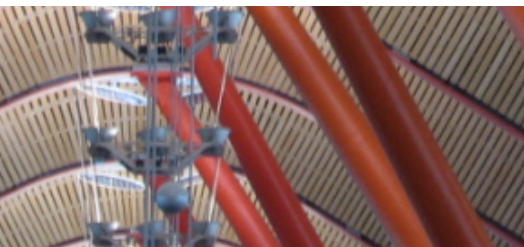
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Since its inception in 2003, the Stanford Electrical Engineering and Computer Science Research Journal (ECJ) has focused on the promotion of research in the Electrical Engineering and Computer Science disciplines. The last two issues of ECJ have featured interesting and cutting-edge academic papers from the Stanford student community, but there is more to research than just science. Scientific research is not just about theories, equations, experiments, and proofs. Behind every scientific discovery or success, there is a person, a story, a community. In this year's issue, we have expanded ECJ to strive to capture this human element of engineering.

In the spirit of giving a voice to the student engineering community at Stanford, the community pieces represent the opinions, passions, advices, and experiences of your fellow students. These pieces convey insights and stories from a quarter abroad, an investigation into Stanford Engineering's history, or the retrospectives of graduating seniors. Regardless of whether their views represent the reader's Stanford experience, these writers are a part of our program and we strongly believe that their thoughts offer great insights into the Stanford EECS community.

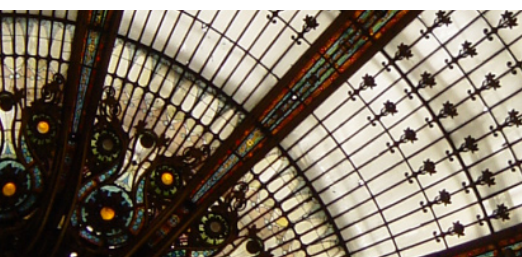
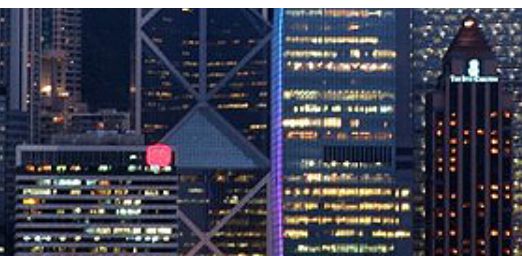
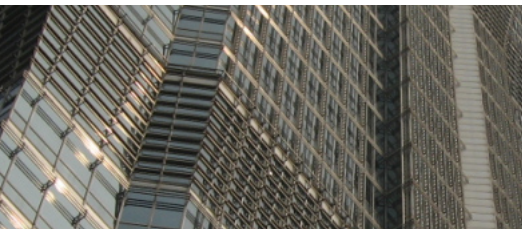
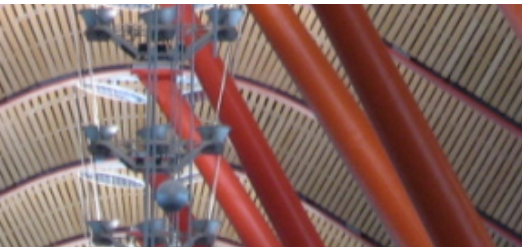
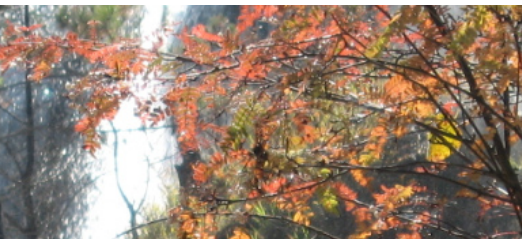
Staying true to ECJ's original mission of promoting research, we have expanded the journal to feature class projects and REU/CURIS reports in addition to independent submissions as in the past. These REU/CURIS projects are the product of an entire summer or longer of hard work by fellow students who have worked closely with a Stanford faculty member. The recognized REU/CURIS projects have been carefully selected by ECJ editors with faculty input, and represent some of the best student work from the summer of 2007. For the upcoming issue of ECJ, we invite all summer 2008 REU and CURIS students to submit reports of their work for publication.

Finally, the ECJ team would like to express its warmest appreciation to our sponsors – the Stanford Electrical Engineering Department and IEEE Santa Clara Valley Council. Their generous support has made this issue of ECJ possible.

We hope that you find the articles in this journal interesting, and more importantly, relevant. We urge the reader to not only take part in research, but to also join in the discussion within our community.

Best,
Ian Wong
Gary Chang

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Stanford EE CS Research Journal (ECJ) is the exclusive student engineering publication on campus. It is produced annually by members of the Stanford chapter of Institute for Electrical and Electronics Engineers (IEEE), a student group on campus. The mission of ECJ is to:

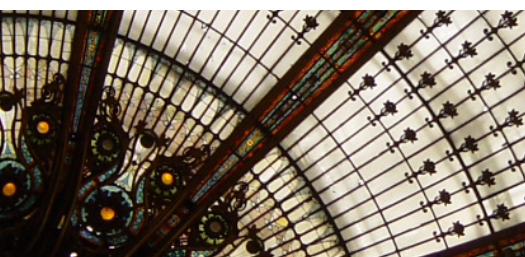
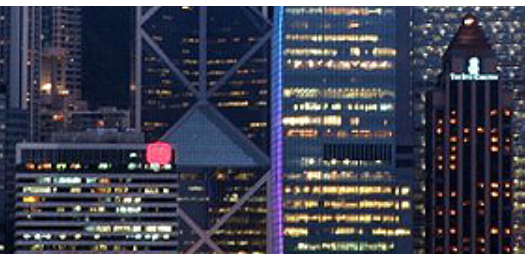
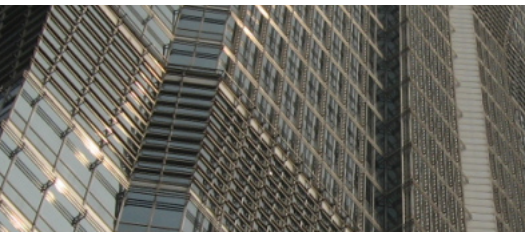
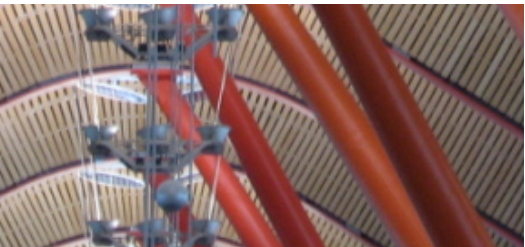
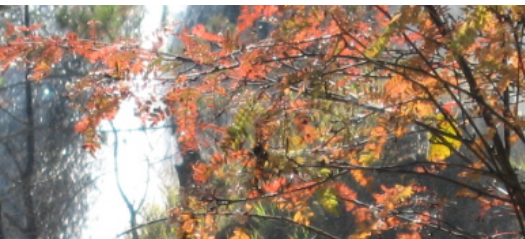
- 1) create awareness across disciplines within the campus technical community,
- 2) prepare current researchers for the publication and review process of larger journals, and
- 3) inspire future research by Stanford community members.

Founded in 2003, ECJ sought to bridge the gap between current undergraduate research presentation opportunities and highly specialized technical journals largely dominated by graduate students, post-doctoral students, and faculty.

In 2007, has undergone substantial changes to make the publication more approachable to a wider set of audience. In the process, the focus of the journal has been expanded to increase the sense of community within the Stanford EE and CS communities. The staff saw great potential for ECJ to be a powerful forum of discussion between all members of the EE CS community. To this end, the staff introduced articles written by ECJ writers in place of long technical papers. While keeping research as the theme of ECJ, the ECJ editor staff felt increasing the scope and flexibility of our articles would augment the journal's relevance and impact on the community and, in turn, better fulfill ECJ's mission.

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One of the great benefits for undergraduates studying in a research university like Stanford, is the opportunity to participate in leading edge research. While not all students will take advantage of this opportunity, for those that do, the results can be magical. All too often an undergraduate engineering or science education can seem like an endless series of lectures, homework assignments and exams. There is much more to engineering and science than this. Working on the frontiers of a discipline by spending a summer in a faculty research group or by doing a special project with a faculty member can provide tremendous insight into what a career might be like. Sometimes these research experiences result in published work, in technical journals or in magazines like this EE/CS Research Journal. I hope you enjoy reading about the research projects of the students who have written for this magazine. Behind each of the articles is a story of an exciting opportunity to explore the frontiers of a field.

Jim Plummer,
Dean, School of Engineering



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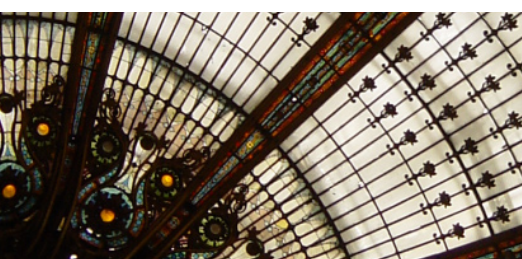
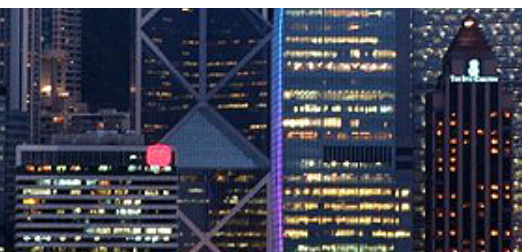
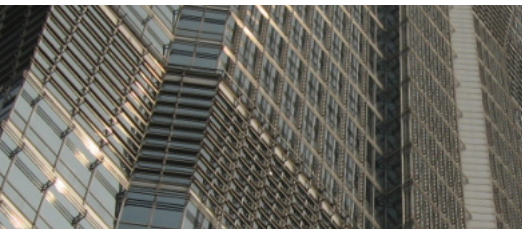
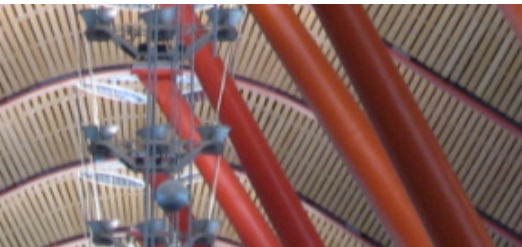
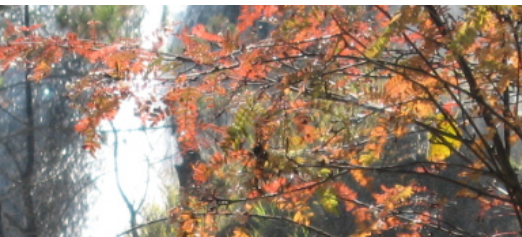
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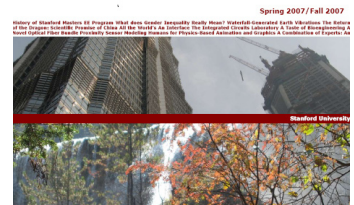
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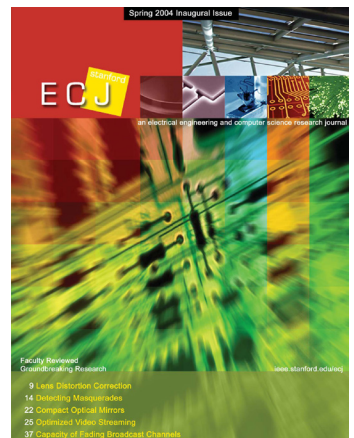
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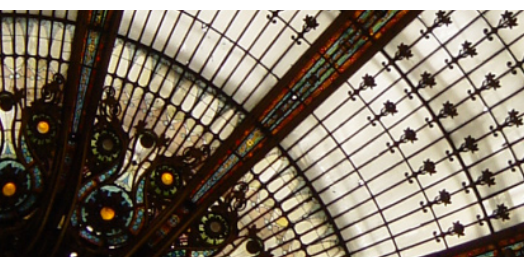
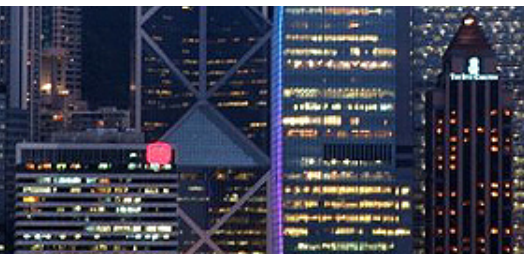
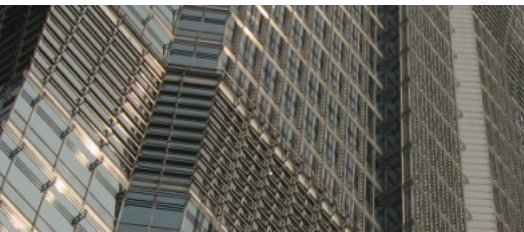
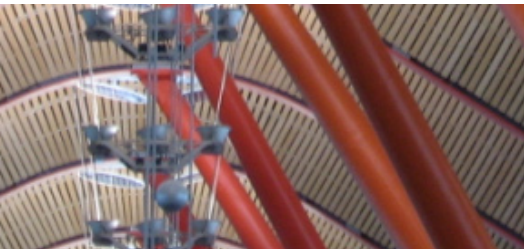
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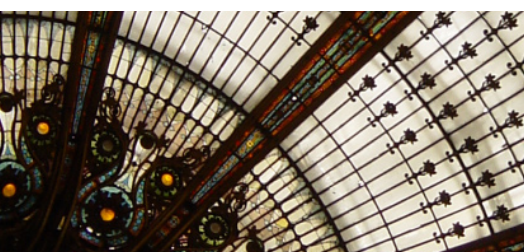
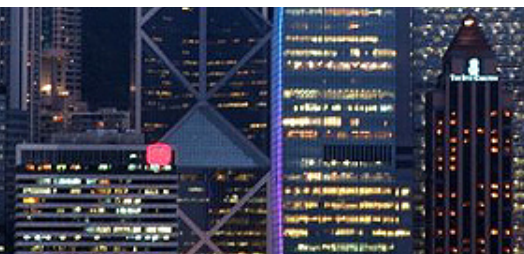
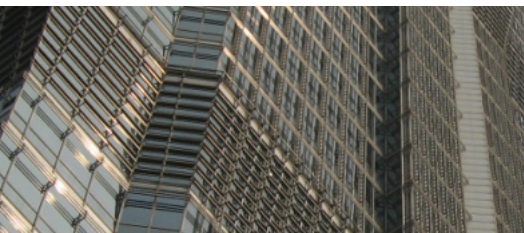
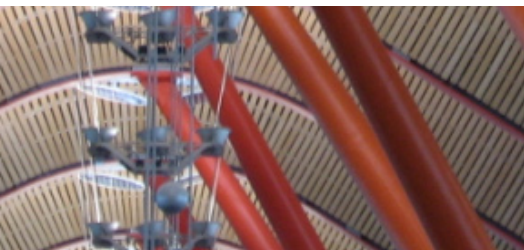
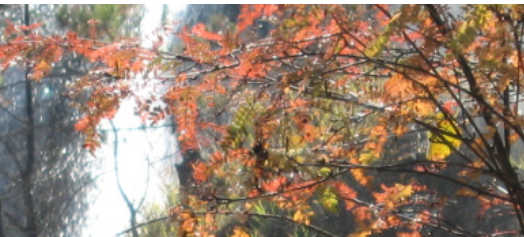
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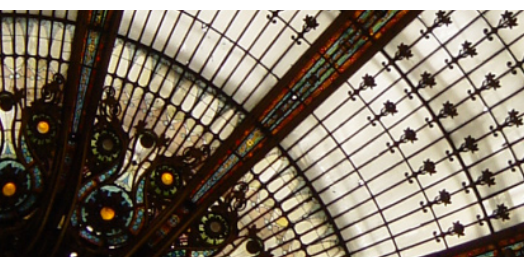
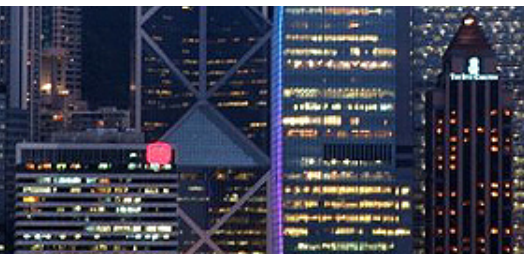
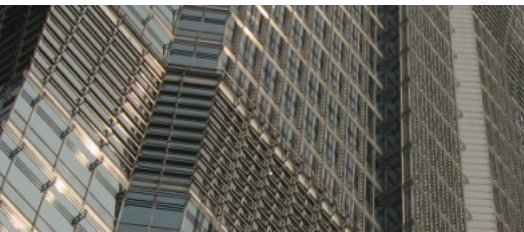
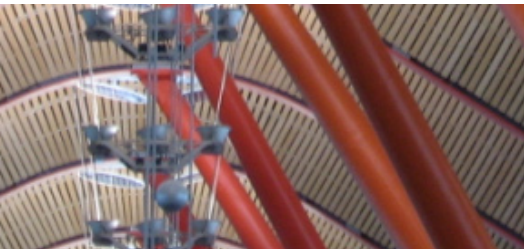
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